Multimedia Signal processing

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1. **Question (8 pts)**:
   Consider two sources, X and Y, that emit symbols according to the following joint p.m.f.

$$\Pr\{X = i, Y = j\} = w_{i,j}, \tag{1}$$

   where the probability values $w_{i,j}$ are defined in the matrix below

$$\mathbf{W} = \frac{1}{8} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2}$$

   (a) Compute the entropy of both sources, i.e., $H(X)$ and $H(Y)$.

   (b) Compute the joint entropy $H(X, Y)$

   (c) Quantize each source with 1 bit. Design the (scalar) Lloyd-Max quantizer. Initialize the intervals as $I_1 = (-\infty, a_2]$, $I_2 = (a_2, +\infty]$, $a_2 = 2$.

   (d) Design the optimal vector quantizer to jointly quantize X and Y, allocating 1 bit per dimension. Initialize the centroids with the reconstruction levels found with Lloyd-Max. Execute only one iteration.

   (e) In both cases, compute the quantization MSE and a lower bound on the minimum number of bits necessary to encode the output symbols of the quantizer.

2. **Question (5 pts)**
   (a) Illustrate the block diagram of a perceptual audio codec (both encoder and decoder). Briefly describe the input and output of each block.

   (b) Describe why adaptive window switching is beneficial in terms of coding efficiency.

3. **Question (5 pts)**
   (a) Illustrate the block diagram of a video coding system.

   (b) Discuss the computational complexity of motion estimation.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Let X denote a continuous source, and $D(R)$ the corresponding rate-distortion function, and σ_x^2 its variance.
   - [T] [F] $H(X) \leq R(D)$
   - [T] [F] If $D(R)$ is achievable, then $D(R - \Delta R)$ is also achievable, when $\Delta R \geq 0$
   - [T] [F] $R(D) \leq \frac{1}{2} \log_2 \frac{\sigma_x^2}{D}$

   (b) Consider a speech coding system
   - [T] [F] Encoding good quality speech requires at least 96 kbps
   - [T] [F] In the case of voiced speech, the output of the short-term predictor is periodic.
   - [T] [F] The perceptual filter attenuates the error components at frequencies corresponding to the formants.

   (c) Let X denote a discrete source with memory.
   - [T] [F] Shannon's coding is such that its average rate $R_S < H(X) + 1$
   - [T] [F] An instantaneous code is always uniquely decodable.
   - [T] [F] The shortest codeword in a Huffman code has always length equal to one.

5. **Question (Matlab - 10pts)**

   Local image features are routinely used in several computer vision applications, including augmented reality (e.g Google Goggles). Their success is due to the ability of summarizing the input image in a way that is invariant to several transformations (scale, rotation, luminance, viewpoint etc.).

   (a) The file `feature_set.mat` contains a set of 64-dimensional SURF local features, extracted from a test image. You are asked to study the performance of DPCM coding on this feature set.
   - Load the file `feature_set`. Implement a closed-loop DPCM coding scheme on the feature set, i.e. quantizing the residuals between adjacent features with R bits per element ($1 \leq R \leq 8$). Plot the resulting rate distortion curve (R [bits per element] vs SNR [dB]).
     Hint: Consider each dimension d of the feature set as a different source. In other words, you have 64 separate DPCM encoders, one per dimension.

   (b) Recently, W. Gao et al. have proposed to sort the feature set in such a way that similar features appear one next the other in the set. This operation clearly improves the DPCM performance, since it reduces the variance of the residuals with respect to the case of the unordered set.
   - Load the file `feature_set_sorted.mat`, that contains the same set of features, this time sorted such that the distance between two consecutive features is minimized. Plot the ordered and the unordered features sets. (Hint: use the `imagesc` function). Compare the performance of the closed-loop DPCM scheme on the ordered set with respect to the unordered case, showing the new rate distortion curve.